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Architecture Comparison of MOLAP, ROLAP, and HOLAP

1. Introduction

Online Analytical Processing (OLAP) is a technology used to analyse large amounts of business data from different perspectives. It helps organisations prepare reports, identify trends, compare performance, and make informed decisions. The three main OLAP architectures are Multidimensional OLAP (MOLAP), Relational OLAP (ROLAP), and Hybrid OLAP (HOLAP). They differ in how they store data, process queries, and provide analytical results.

2. Detailed Comparison Table

Feature	MOLAP	ROLAP	HOLAP
Full meaning	Multidimensional OLAP	Relational OLAP	Hybrid OLAP
Data storage	Stores data in multidimensional cubes.	Stores data in relational database tables.	Stores summaries in multidimensional cubes and detailed data in relational tables.
Query performance	Generally very fast because data and aggregations can be preprocessed.	May be slower because queries often access relational tables.	Fast for summary queries, while detailed queries depend on the relational database.
Data processing	Data is processed into cubes before analysis.	Queries are processed against relational data, commonly using SQL.	Combines multidimensional processing with relational database queries.
Storage requirements	Requires storage for cubes and their aggregations.	Uses the relational database for storing data.	Requires both relational storage and multidimensional storage.
Scalability	Can face storage and processing challenges as cubes grow.	Can handle very large datasets when supported by a suitable database.	Balances multidimensional performance with relational data capacity.
Data updates	Changes may require cube processing or refreshing.	Can access updated source data more directly, depending on configuration.	Requires management of cube summaries and relational source data.

Main advantage	Provides fast access to frequently requested analytical information.	Supports analysis of large amounts of detailed data.	Combines fast summaries with access to detailed records.
Main disadvantage	Cube processing can take time and consume additional storage.	Complex analytical queries may take longer to execute.	More complex to configure and maintain.
Common application	Sales analysis, budgeting, and financial reporting.	Enterprise data warehouses and large-scale business reporting.	Business intelligence systems requiring both fast summaries and detailed analysis.

3. Research on Three Commercial OLAP Tools

3.1 Microsoft SQL Server Analysis Services (SSAS)

Microsoft SQL Server Analysis Services is a business intelligence platform used to build analytical models and multidimensional cubes. Its Multidimensional mode supports MOLAP, ROLAP, and HOLAP storage options.

MOLAP is commonly selected when fast query performance is a priority because data and aggregations can be stored in multidimensional structures. ROLAP can be selected when direct access to relational data is required, while HOLAP stores aggregations in multidimensional structures and detailed data in relational tables.

Therefore, Microsoft SSAS supports all three architectures, and the primary architecture depends on the storage configuration chosen by the organisation.

Reference: Microsoft Learn, *Set Partition Storage in Analysis Services*. https://learn.microsoft.com/en-us/analysis-services/multidimensional-models/set-partition-storage-analysis-services-multidimensional

3.2 Oracle OLAP

Oracle OLAP is an analytical technology integrated into the Oracle Database environment. It uses analytic workspaces to store multidimensional data, dimensions, calculations, and aggregations.

Oracle OLAP is primarily associated with MOLAP-style multidimensional storage because its analytic workspaces are designed to support multidimensional calculations and fast analytical queries. However, these workspaces are integrated into the Oracle relational database, combining multidimensional analytical capabilities with database management features.

This architecture is useful for organisations that need to perform complex calculations, analyse historical data, and generate business summaries.

Reference: Oracle, *Oracle OLAP User's Guide*. https://docs.oracle.com/database/121/OLAUG/toc.htm

3.3 IBM Cognos Analytics

IBM Cognos is a business intelligence platform used for reporting, dashboards, and data analysis. Its OLAP architecture depends on the specific component being used.

IBM Cognos Dynamic Cubes primarily uses a ROLAP-based approach with in-memory acceleration. It creates a multidimensional view of data stored in relational data warehouses and uses caching and optimised queries to improve performance.

IBM also provides other OLAP technologies. For example, Cognos TM1, now associated with IBM Planning Analytics, uses in-memory MOLAP technology. Therefore, it is important to identify the specific Cognos component when describing its architecture.

Reference: IBM, *Cognos Dynamic Cubes Overview*. https://www.ibm.com/docs/en/cognos-analytics/11.2.x?topic=cubes-cognos-dynamic-overview

4. Summary of Commercial OLAP Tools

Commercial tool	Primary architecture or approach	Explanation
Microsoft SQL Server Analysis Services	MOLAP commonly used; supports MOLAP, ROLAP, and HOLAP	Offers several storage modes to meet different performance and data-access requirements.
Oracle OLAP	MOLAP-style	Uses multidimensional analytic workspaces integrated into Oracle Database.
IBM Cognos Dynamic Cubes	ROLAP with in-memory acceleration	Analyses relational warehouse data using caching, optimised SQL, and in-memory aggregates.

5. Conclusion

MOLAP, ROLAP, and HOLAP are important OLAP architectures that support business data analysis in different ways. MOLAP provides fast analytical queries through multidimensional storage and precomputed aggregations. ROLAP works directly with relational database tables and is suitable for analysing large volumes of detailed data. HOLAP combines the advantages of both approaches by storing summary information in multidimensional cubes while retaining detailed data in relational databases.

The research shows that Microsoft SSAS supports all three architectures, Oracle OLAP primarily uses multidimensional analytic workspaces, and IBM Cognos Dynamic Cubes uses a ROLAP-

based approach with in-memory acceleration. The most suitable architecture depends on an organisation's data volume, performance requirements, storage capacity, and need for up-to-date information.

2. Architecture Comparison: MOLAP, ROLAP and HOLAP

1. Comparison of OLAP Architectures

Online Analytical Processing (OLAP) is a technology used to analyze large amounts of data from different perspectives to support business decision-making. The three main OLAP architectures are MOLAP, ROLAP and HOLAP. They differ in how they store data, process queries and manage large datasets.

TechTarget
+1

Feature	MOLAP	ROLAP	HOLAP
Full name	Multidimensional OLAP	Relational OLAP	Hybrid OLAP
Data storage	Stores data in multidimensional cubes.	Stores data in relational database tables.	Stores detailed data in relational tables and summaries in multidimensional cubes.
Data processing	Data is processed and aggregated before queries are executed.	Data is processed using SQL queries against relational databases.	Combines precomputed cube aggregations with relational data processing.
Query performance	Very fast for precomputed summaries and cube queries.	Can be slower for complex queries involving large tables and joins.	Fast for summaries, but detailed queries may be slower.
Storage requirements	May require additional storage for cubes and precomputed aggregations.	Uses existing relational database storage, with additional space for indexes and summaries.	Requires both relational storage and multidimensional storage.
Scalability	Suitable for structured analytical workloads; cube size and processing can become limitations.	Well suited to very large datasets stored in relational databases.	Combines relational scalability with faster access to summarized data.

Data updates	Often requires cube processing or refreshing to reflect changes.	Can provide access to recent data directly from the relational database.	Detailed data can remain current while cube summaries may need refreshing.
Maintenance	Requires cube processing, aggregation design and storage management.	Requires database tuning, indexing and query optimization.	Requires management of both relational data and cube aggregations.
Advantages	Fast queries, efficient multidimensional calculations and interactive analysis.	Handles large datasets and supports access to detailed relational data.	Balances query speed, scalability and detailed-data access.
Disadvantages	Cube processing can take time, and storage requirements can increase.	Complex queries may take longer to execute.	More complex to configure and maintain because it combines two storage approaches.
Best use case	Sales dashboards, budgeting and financial analysis with frequent summary queries.	Large enterprise databases, detailed transaction analysis and historical reporting.	Business intelligence systems that need fast summaries and access to detailed records.

Source: IBM's OLAP overview, TechTarget's architecture comparison, and Microsoft Learn documentation.

IBM
+2

2. Research on Three Commercial OLAP Tools

Three commercial OLAP tools and their primary architectures are compared below.

Commercial tool	Primary OLAP architecture	Explanation and uses
Microsoft SQL Server Analysis Services (SSAS)	MOLAP by default; also supports ROLAP and HOLAP	SSAS multidimensional models use MOLAP as the default storage mode. Administrators can select ROLAP for relational storage or HOLAP to combine relational detail data with multidimensional aggregations. It is used for enterprise reporting, business intelligence and data analysis.

Microsoft Learn
+1

Oracle Essbase	MOLAP	Oracle Essbase organizes business data into multidimensional cubes. It supports financial planning, budgeting, forecasting and analysis of business performance. Oracle Docs +1
IBM Cognos TM1 (Planning Analytics)	MOLAP	TM1 uses an in-memory multidimensional database engine and stores business data in cubes. It is mainly used for financial planning, budgeting, forecasting and what-if analysis. IBM +1

Note: IBM Cognos is a product family with different OLAP technologies. Cognos TM1 primarily uses MOLAP, while Cognos Dynamic Cubes uses an in-memory ROLAP approach with caching and aggregate acceleration. Therefore, the architecture depends on the specific Cognos component being used.

IBM
+1

3. Conclusion

MOLAP, ROLAP and HOLAP each provide different advantages. MOLAP is suitable when fast analytical queries are required, ROLAP is useful for analyzing very large relational datasets, and HOLAP combines both approaches to provide fast summaries and access to detailed records.

The research shows that Microsoft SSAS supports all three architectures, while Oracle Essbase and IBM Cognos TM1 primarily use MOLAP. The best architecture depends on the size of the daset, query performance requirements, data freshness and available storage resources

3.Performance Analysis Case Study: OLAP Implementation at Xbox LIVE

Introduction

Online Analytical Processing (OLAP) is used by organizations to analyze large amounts of data and support decision-making. A real-world example is Microsoft's Xbox LIVE data warehousing environment, which used SQL Server Analysis Services (SSAS) to perform analytical operations. The implementation faced performance challenges when processing large datasets, especially when calculating distinct counts, such as the number of unique users or customers.

Performance Challenges

One major challenge was the time required to execute complex queries. Distinct-count calculations are more demanding than simple calculations because the system must identify unique values instead of merely adding numbers. When users requested reports involving large datasets and several dimensions, the system required more processing power, memory and storage access.

Another challenge was maintaining fast response times as data volumes and analytical requirements increased. If a cube was poorly designed or lacked suitable aggregations, the system could be forced to process large amounts of detailed data every time a user requested a report. This could increase query response times and affect the overall performance of the business intelligence system.

Solutions to the Performance Problems

Microsoft investigated the use of solid-state drives (SSDs) and optimized multidimensional OLAP cube designs to improve distinct-count performance. SSDs provide faster data access than traditional hard disk drives, reducing delays caused by reading data from storage.

Cube design was another important factor. Organizing data into appropriate dimensions and measures helps the OLAP engine retrieve information more efficiently. For example, a sales cube can organize data by product, customer, location and time, allowing users to analyze sales from different perspectives.

Aggregation strategies can also improve performance. Aggregations are precomputed summaries of detailed data. Instead of calculating totals from thousands of individual records every time, the system can retrieve previously calculated summaries. For example, monthly sales totals can be used to calculate quarterly sales more quickly. Administrators can prioritize aggregations based on the queries users run most frequently.

In addition, monitoring query performance and allocating sufficient memory and processing resources help administrators identify bottlenecks and improve the system's efficiency.

Conclusion

The Xbox LIVE case study demonstrates that OLAP performance depends on effective cube design, suitable aggregation strategies and appropriate hardware. Microsoft's investigation of SSD storage and optimized multidimensional cubes addressed demanding analytical workloads. This case shows that organizations should combine software optimization with suitable hardware resources to reduce query response times and improve scalability. These improvements enable businesses to analyze information more efficiently and make better decisions.

References

1. Microsoft SQL Server Analysis Services. *Distinct Count Optimization Using Solid State Devices: A Real Practices Case Study at Xbox LIVE.* Available at: https://www.ssas-info.com/analysis-services-articles/61-performance-tuning/2423-analysis-services-distinct-count-optimization-using-solid-state-devices
2. Microsoft. *SQL Server Analysis Services Performance and Operations Guide.* Available at: https://learn.microsoft.com/en-us/analysis-services/